

Francis Lufwendo

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EDUCATION

Yeshiva University, The Katz School of Science and Health

New York, USA

Master of Science in Biotechnology Management and Entrepreneurship; GPA: 3.88

Present

Relevant Coursework: Venture Creation and entrepreneurship, Project Management, Biotech management, Foundations of Biotechnology/ Molecular Biology, Bioinformatic, CRISPR & Gene Editing Technologies,

Marwadi University

Gujarat, India

Bachelor of Science in Bachelor of Science in Pharmacy; GPA: 3.77;

May 2024

Senior Thesis: "Engineering a Polyherbal Emulsion for Septicemia Survival"

Publication: Co-authored chapter 'Artificial Intelligence and Machine Learning in Drug Discovery' in *Modern Approaches in Drug Discovery and Development*

Foundations of AI Engineering, CodePath

New York, USA

Coursework: RAG, agentic workflows, ML literacy, fine-tuning, guardrails AI-assisted programming, GitHub & Open-Source Collaboration

TECHNICAL SKILLS

Programming Languages: JavaScript, Python, SQL, HTML/CSS

Frameworks & Libraries: React, Node.js, Express.js, MongoDB (MERN Stack), TensorFlow/PyTorch, MySQL, PostgreSQL, OpenAI API, Hugging Face,

AI/ML: Retrieval-Augmented Generation (RAG), Natural Language Processing, Large Language Models, Langchain

Data Analysis and Research: Tableau, R-language, GraphPad Prism, Qlik BioEdit, EMBL-EBI tools, PubMed, Google scholar Pandas, NumPy, Matplotlib

Tools & Platforms: Git/GitHub, RESTful APIs, AWS/Cloud Services

Domain Expertise: FDA Regulatory Affairs, Clinical Research, Pharmaceutical Sciences, Biotech R&D Workflow

WORK EXPERIENCE

RegComplyAI - AI-Powered FDA Compliance Platform

New York, USA

| Founder & Lead Developer |

Present

- Architected and developed **full-stack web application** using MERN stack to streamline FDA regulatory compliance for early-stage biotech companies, targeting \$16.4B regulatory affairs market
- Implemented **Retrieval-Augmented Generation (RAG)** system to parse and extract insights from 500+ FDA regulatory documents, reducing compliance research time by 60%
- Designed and built **RESTful API** endpoints to handle **complex scientific workflow automation**, including document management, **submission tracking**, and **regulatory timeline visualization**
- Managed end-to-end product development from **market research** and **technical architecture** to **prototype deployment**, incorporating user feedback from biotech founders
- Integrated Python-based document processing pipeline** with **React frontend**, troubleshooting dependency conflicts and optimizing **cross-platform** compatibility
- Technologies:** React, Node.js, Express, MongoDB, Python, RAG/LLM APIs, Git/GitHub
Website: <https://regcomplyai.org>

Education-24: Non-profit online college access program

Lusaka, Zambia

| Co-founder and Mentor |

Sep 2021 – Present

- Co-founded and led an initiative providing free college access support to high-achieving, low-income students in Southern Africa.
- Designed and developed the **website** for the program.
- Established partnerships with educational institutions in the USA to expand program reach.
- Education24 has secured over **\$2 million in scholarships** for our students,
- Website: <https://www.education-24.com>

Professional Development

- Active participant in **Harvard Biotech Club** webinars and **NVIDIA** tech meetups
- Attend Wellie Cornell event, and Rockefeller University
- Building professional network across biotech and tech industries in the Northeast corridor

Languages: English (fluent), Tonga and Nyanja (native), Swahili (basics), Hindi (basics)
Honors: Stem Fellows Scholarship 2025, Government of India Scholarship Recipient 2021
